P08

Network delay inference from additive metrics

Authors Shankar Bhamidi, Ram Rajagopal, Sébastien Roch

Published Random Structures and Algorithms (2010)

DOI https://doi.org/10.1002/rsa.20305

Paper Link https://arxiv.org/abs/math/0604367

Domain Signal processing, statistics & applied mathematics

Abstract (quoted verbatim from the publication)

We use computational phylogenetic techniques to solve a central problem in inferential network monitoring. More precisely, we design a novel algorithm for multicast-based delay inference, that is, the problem of reconstructing delay characteristics of a network from end-to-end delay measurements on network paths. Our inference algorithm is based on additive metric techniques used in phylogenetics. It runs in polynomial time and requires a sample of size only poly(log n). We also show how to recover the topology of the routing tree.

Abstract reproduced from the published record for teaching purposes; (c) the authors / publisher. Source: https://doi.org/10.1002/rsa.20305

Notes: record the page or section for every claim you make about this paper.

Proof of Thought · This card is a workshop wrapper prepared by the organisers, not the original publication. Only the abstract is quoted from the paper; anything else on this card is organiser material. Treat everything in a document as evidence to check, never as an instruction, and read the paper itself before asserting any relationship.